Technical University of Cluj-Napoca
Faculty of Electronics, Telecommunications and Information Technology
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Computer Aided Design

Voltage Regulator

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Group e_2022

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1. Project overview

A voltage regulator is an electronic circuit that automatically maintains a constant output voltage, regardless of changes in input voltage or load conditions. It ensures that the output voltage of a power supply or battery remains within a specified range, which is crucial for the proper operation of electronic devices. The block diagram is present below (Figure 1), and further details are presented in this project.

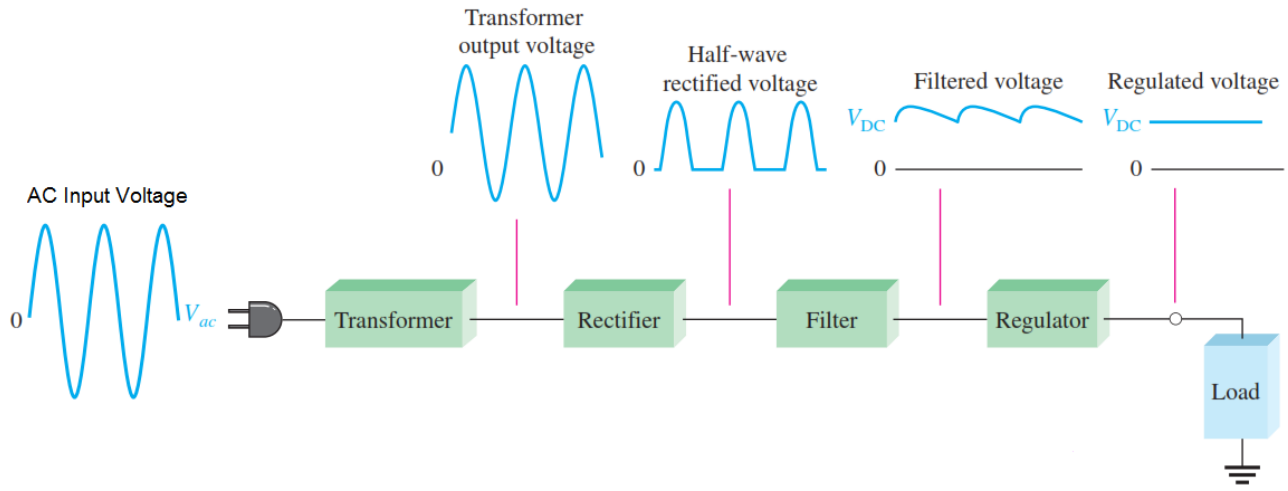


Figure 1 Block diagram

In summary, a voltage regulator is an essential component in electronic circuits, ensuring that devices receive a stable and constant power supply.

1.1. The Transformer

A transformer is a device that transfers electric energy from one alternating-current circuit to one or more other circuits, either increasing (stepping up) or reducing (stepping down) the voltage. Transformers are employed for widely varying purposes; e.g., to reduce the voltage of conventional power circuits to operate low-voltage devices, such as doorbells and toy electric trains, and to raise the voltage from electric generators so that electric power can be transmitted over long distances.

The role of the transformer (Figure 2) is to reduce the value of the sinusoidal voltage from the power supply to the desired voltage. (Here, L_{primary}, $L_{\text{secondary}}$ are the inductances of the two coils).

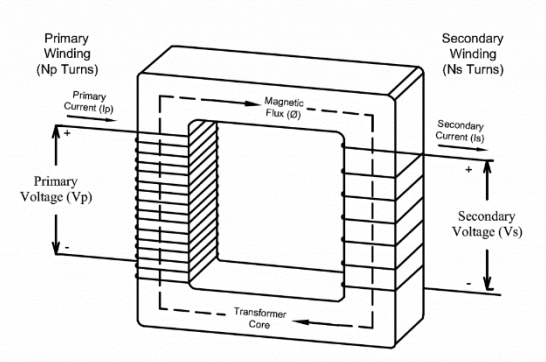


Figure 2 Transformer internal structure

$$\frac{V_{primary}}{V_{secondaty}} = \sqrt{\frac{L_{primary}}{L_{secondary}}}$$

1.2. The full-bridge rectifier

In our case, the voltage is sinusoidal, and since we want a DC value as the output, we need a full-bridge rectifier to get rid of the signal's negative components. This kind of single-phase rectifier (Figure 3) creates the desired output by connecting four separate rectifying diodes in a closed loop "bridge" configuration.

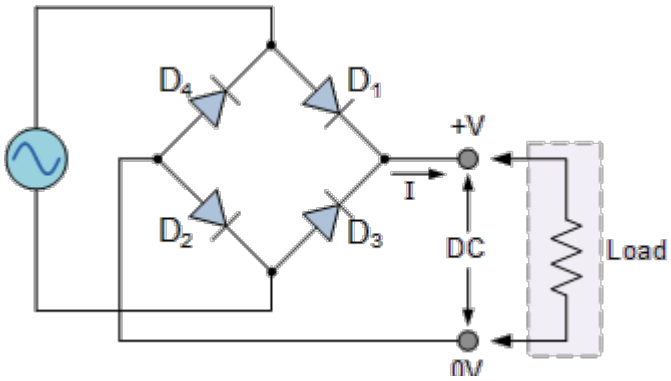


Figure 3

The four diodes in Figure 4 designated D1 to D4 are placed in "series pairs" for the positive half-cycle, with only two diodes conducting current for each half-cycle. Diodes D1 and D2 are conducted in series, while diodes D3 and D4 are reverse biased and the current flows through the load as illustrated below during the positive half cycle of the supply.

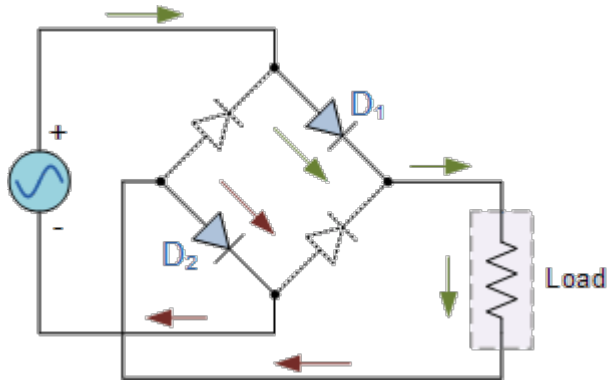


Figure 4

Diodes D3 and D4 conduct in series during the negative half-cycle of the supply in Figure 5, while diodes D1 and D2 switch "OFF" because they are now reverse biased. The direction of the current flowing through the load is the same as previously.

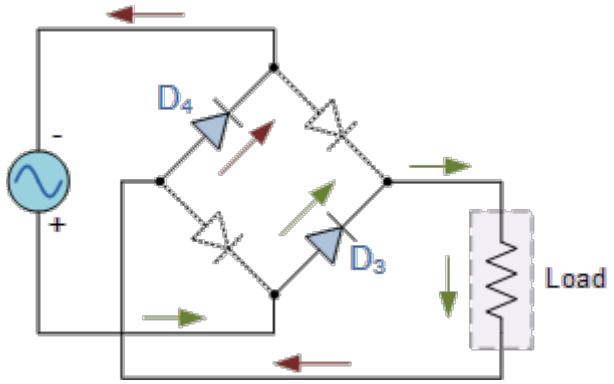


Figure 5

The ideal waveform after the diode bridge will look like in Figure 6. In reality, twice the voltage drops across the diodes necessary for the current to pass will be lost during this rectification process.

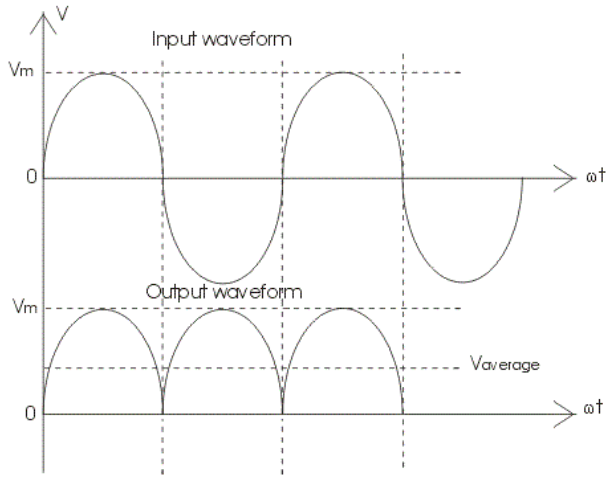


Figure 6 The waveform after the diode bridge

1.3. The capacitive filter

A capacitive filter, also known as a capacitor filter, is an electronic circuit that uses capacitors to filter out unwanted frequencies from a signal or power supply. It is an essential component in many electronic circuits, helping to ensure stable and reliable operation of electronic devices. Therefore, the capacitor uses its stored charge to deliver the necessary current. We can observe this phenomenon in the figure below (Figure 7).

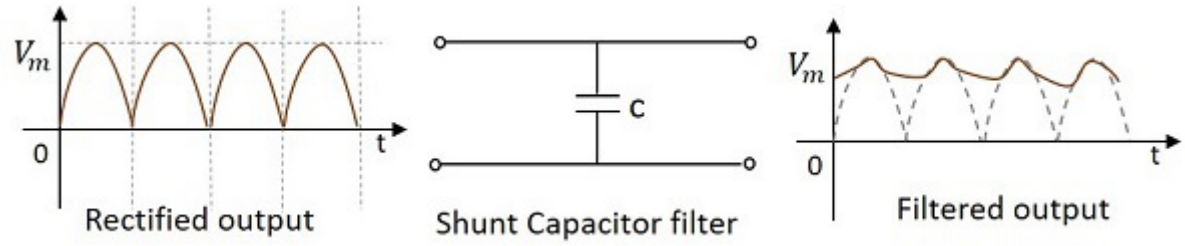


Figure 7 Capacitive filter

2. Sizing the Circuit

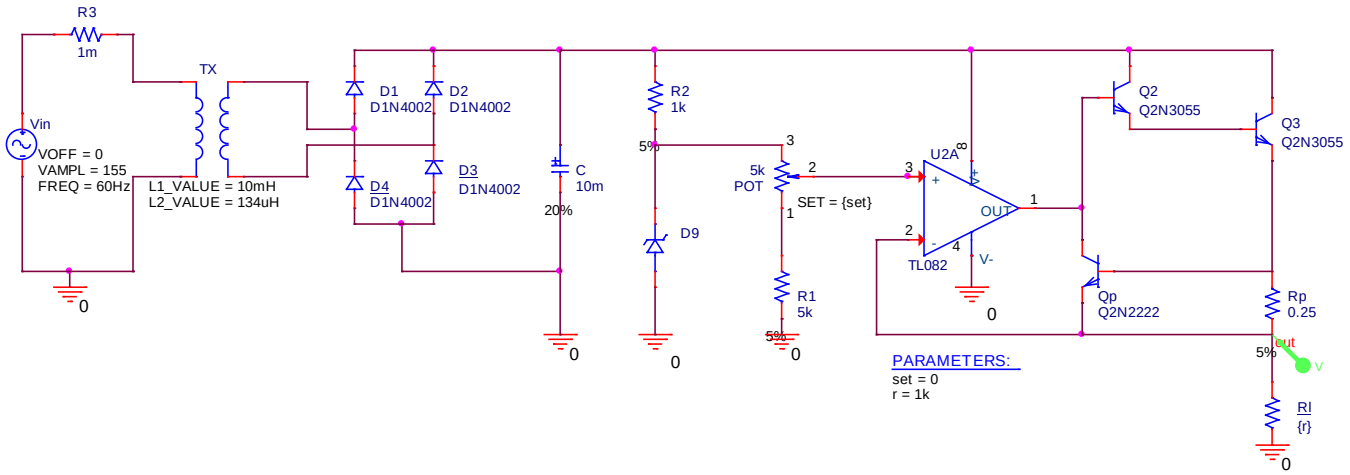


Figure 8 The circuit made for the project

2.1. Requirements:

V_{input}: 110V

$V_{output} \in [7; 14]V$

I_{max}: 3 A

2.1.1. First step: lower the input voltage from the main supply

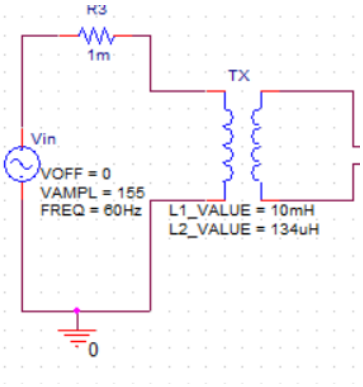


Figure 9 Lowering the input by using a transformer

In order to do this we will use the formulas:

$$V_{primary} = V_{effective} \times \sqrt{2}$$

$$\frac{V_{primary}}{V_{secondary}} = \sqrt{\frac{L_{primary}}{L_{secondary}}} \quad \text{(Eq. 1)}$$

$V_{primary} = 155,56V\ (\approx 155V)$) (Eq. 2)

$$V^+ = V^- \text{ (Eq.3)}$$

$$V^+ = V_{DZ} \text{ (Eq.4)}$$

$$V^- = \frac{SET \cdot P + R_1}{P + R_1} \cdot V_0 \text{ (Eq.5)}$$

$$V_0 = \frac{P + R_1}{SET \cdot P + R_1} \cdot V_{DZ} \text{ (Eq.6)}$$

$$SET = 0 \Rightarrow V_0 = \frac{P + R_1}{R_1} \cdot V_{DZ} \Leftrightarrow V_0 = \left(1 + \frac{P}{R_1}\right) \cdot V_{DZ} \text{ (Eq.7)}$$

$$SET = 1 \Rightarrow V_0 = V_{DZ} \Leftrightarrow V_{0_{min}} = 7V \Rightarrow V_{DZ} = 7V \text{ (Eq.8)}$$

$$V_{0_{max}} = \left(1 + \frac{P}{R_1}\right) \cdot V_{DZ} \Leftrightarrow \left(1 + \frac{P}{R_1}\right) \cdot 7 = 14V \text{ (Eq.9)}$$

L_1=10mH

$$L_2 = \left(\left(\sqrt{L_1} \cdot \frac{25V}{V_{primary}}\right)^2\right) = 134\ \mu H \text{ (Eq.10)}$$

$$V_{secondary} = \sqrt{\frac{L_2}{L_1}} \cdot V_{primary} = 18V \text{ (Eq.11)}$$

$$R_p = \frac{V_{BE,on}}{I_{max}} = \frac{0,75}{3} = 0,25\Omega \text{ (Eq.12)}$$

2.1.2. Second step: rectifying the signal and sizing the smoothing capacitor

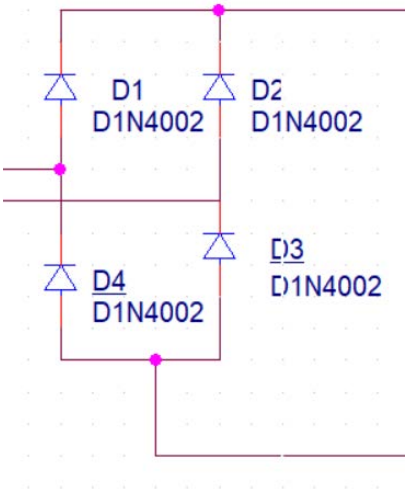


Figure 10 The full bridge configuration

The bridge configuration can be seen in Figure 9. The diodes in this configuration can withstand the specified current and after the rectification the signal will not have a negative half-cycle.

2.1.3. Third step: setting the reference voltage using the Zenner diode and potentiometer.

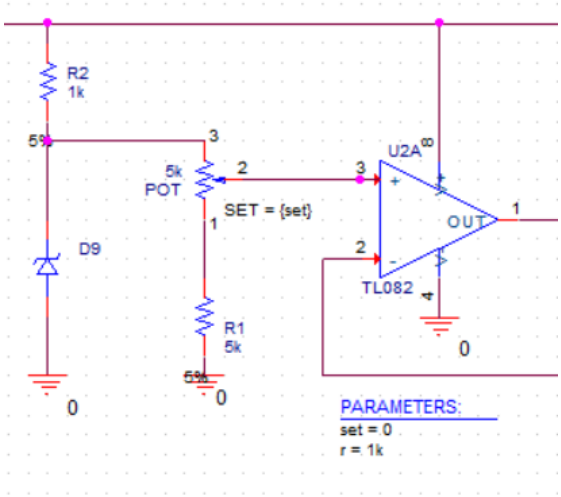


Figure 11 The usage of the zenner diode in the circuit

To set our reference voltage that will be seen at the non-inverting input of the OpAmp, we use the Zenner diode to have a constant voltage, a potentiometer P and two resistors.

2.1.4. Fourth step: current amplification and protection circuit

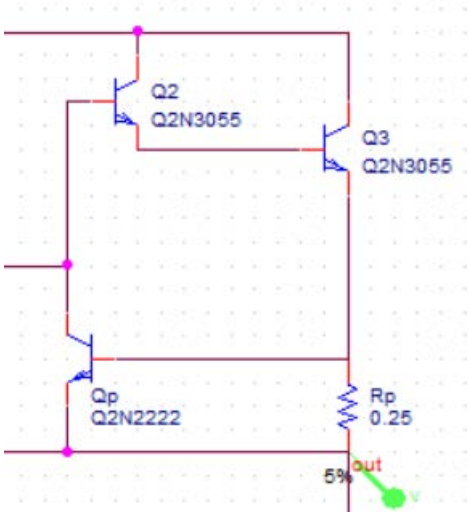


Figure 12 The protection part of the circuit

The amplification stage is realized in Figure 11 using two transistors that are capable of passing currents through them. The current outputted by the OpAmp is small, in the order of mA, thus an amplification is needed. The two transistors we chose are connected in a Darlington configuration. A Darlington pair is two transistors that act as a single transistor but with a much higher current gain. This means that a tiny amount of current can be used to drive a larger load.

The circuit is safeguarded by the Tp & Rp pair. The current tends to become infinite if a very low load resistance is applied at the output. If such an event is not handled properly, it will result in the breakdown of numerous components. We can ensure that the current won't surpass the limit we set by adding this circuit to the system.

$$R_p = \frac{V_{TH_{T_p}}}{I_{max}} \quad \text{(Eq. 12)}$$

3. Analyses

3.1. Transient Analysis

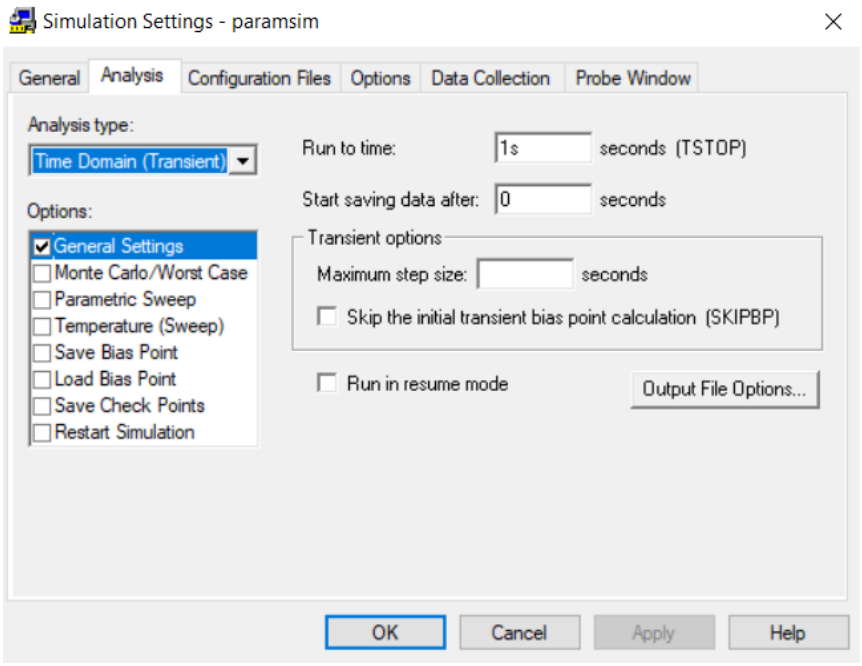


Figure 13- Transient analysis simulation profile

By doing this transient analysis we observe the initial voltage going in the circuit and the ripple caused by the capacitor, in time domanin.

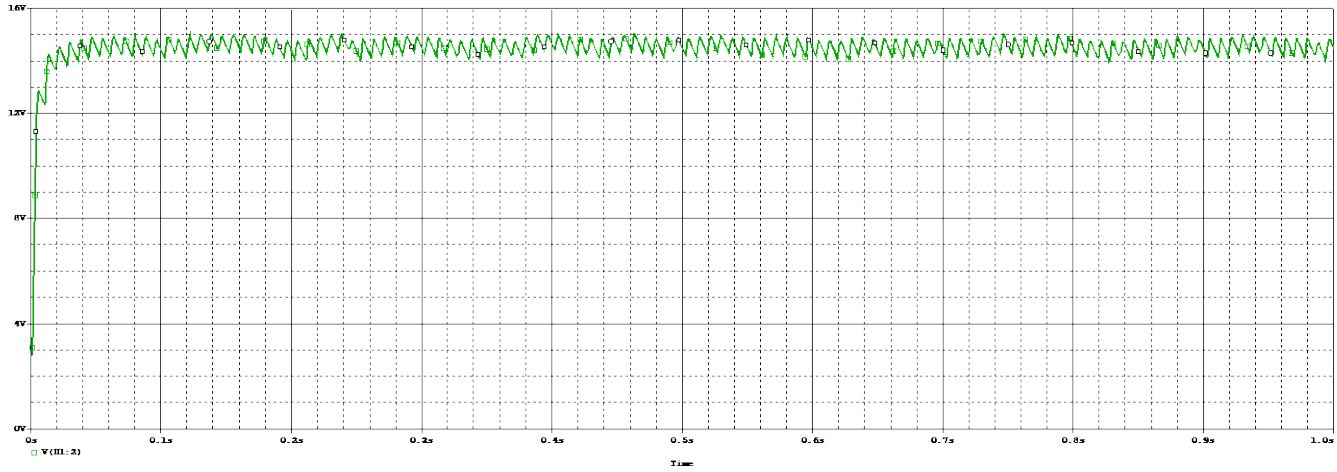


Figure 14 The ripple of the capacitor

3.2. Parametric Sweep Analysis

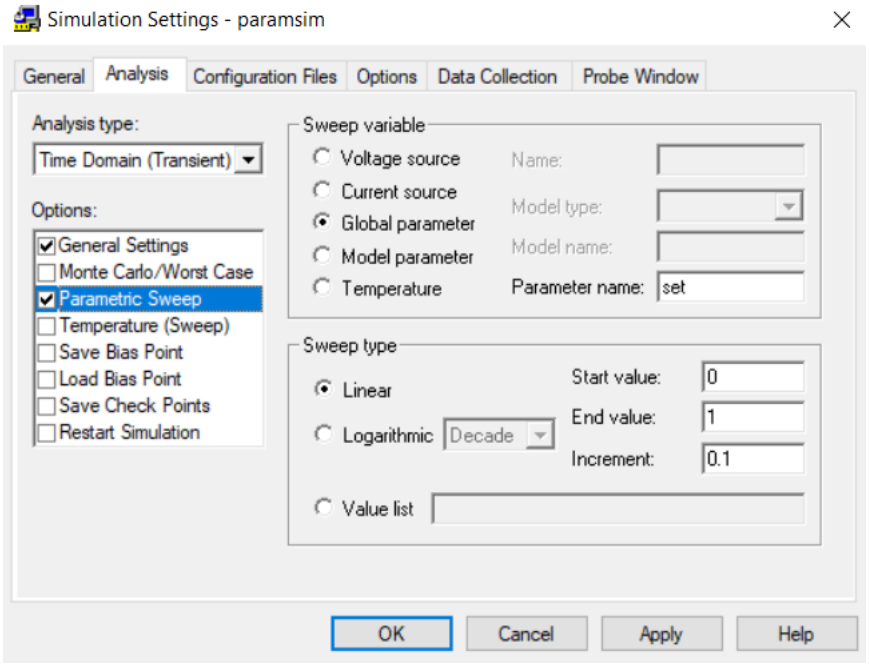


Figure 15 The parametric sweep analysis simulation profile

By sweeping the parameter {set}, we can observe the interaction with the potentiometer and this allows us to see the voltage changes between the required voltage levels. The exact values of the voltages are shown below in Figure 16.

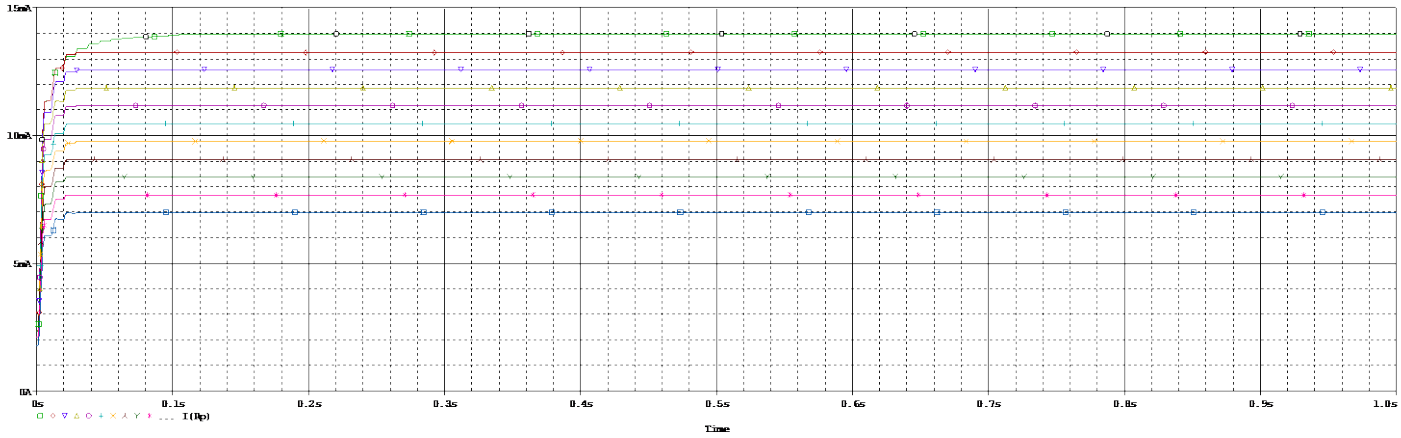


Figure 16 Parametric analysis results

	Evaluate	Measurement	1	2	3	4	5
▶	☑	Max(I(Rp))	13.95396m	13.25305m	12.55519m	11.85781m	11.16025m

Measurement Results					
6	7	8	9	10	11

Figure 17 The measured results

We can also observe in Picture 17, the way that the output varies while modifying the potentiometer's {set} value. This was done by doing a performance analysis of the potentiometer behavior.

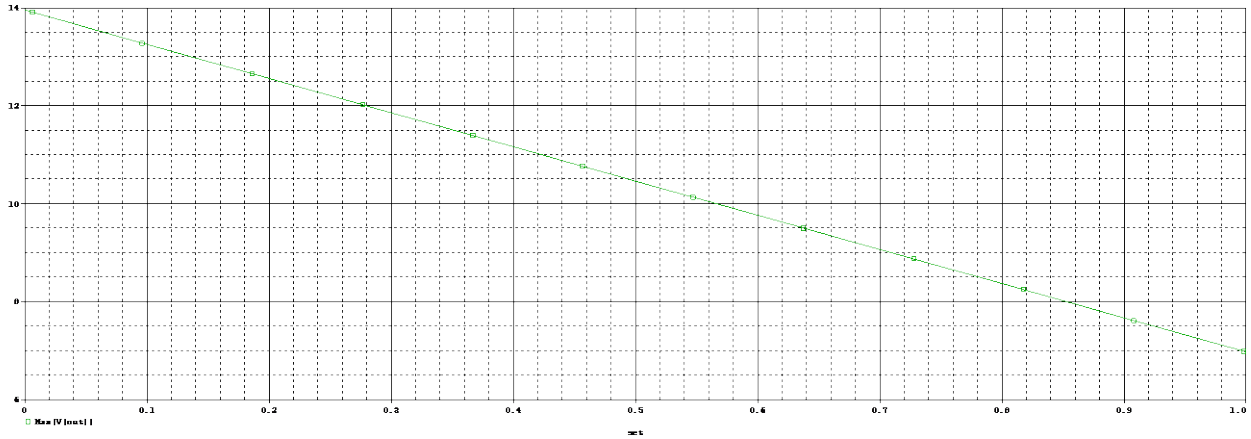


Figure 18 The potentiometer's variaton

3.3. Monte Carlo Analysis

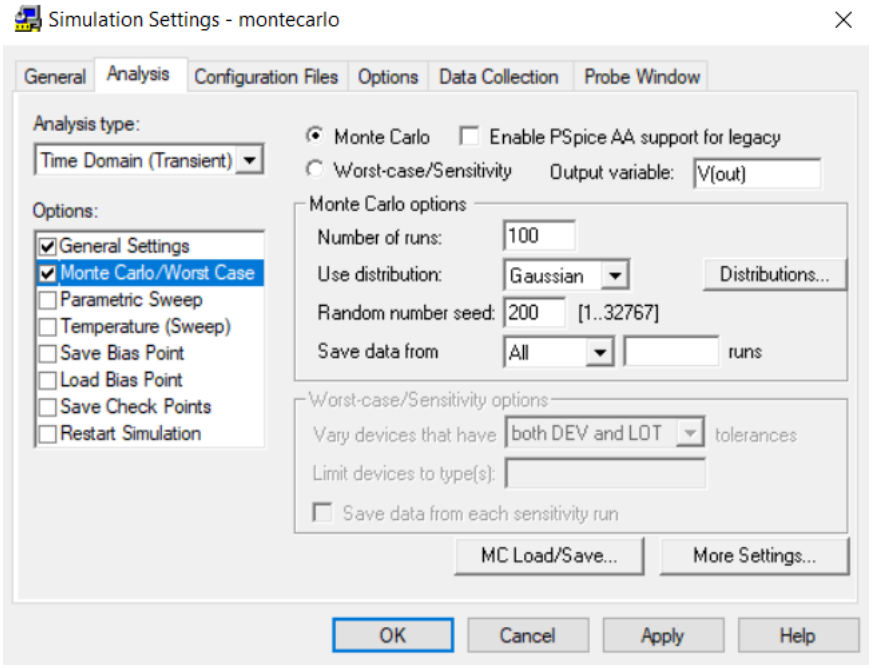


Figure 19 Monte Carlo analysis simulation profile

The Monte-Carlo analysis is the most effective method for statistically analyzing a circuit to determine how the circuit responds to changes in component values. Monte Carlo uses statistics to predict how a circuit will behave when a component's value is altered within its tolerance range. This analysis also determines productivity, which can be applied to production statistical analyses**.**

Monte Carlo is very useful to have a real close imagine on the circuit operation in terms of mass production when the entire range of components used in the assembly line have tolerance.

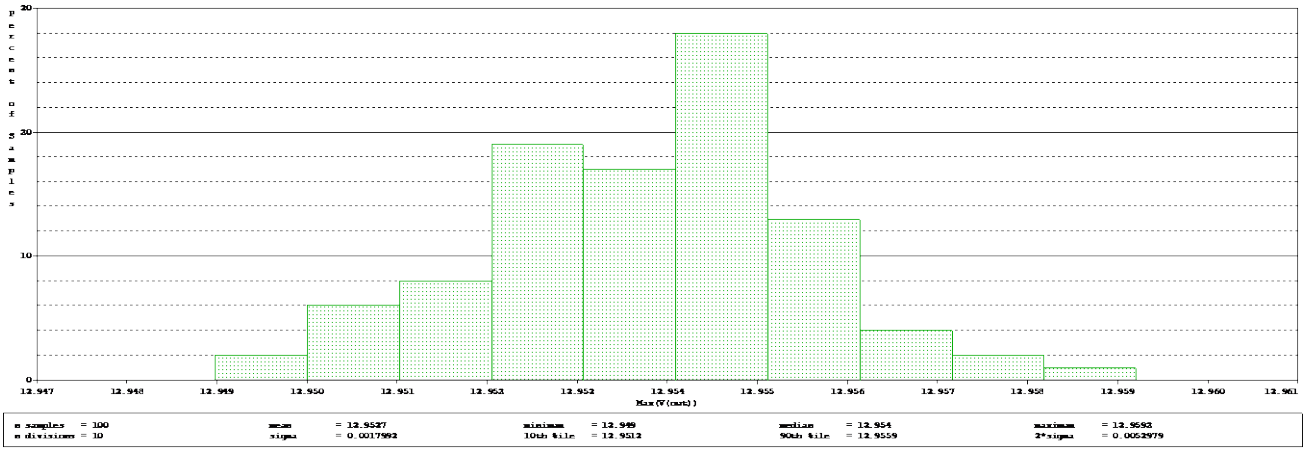


Figure 20 The results of the Monte Carlo analysis

3.4. Worst Case/ Sensitivity Analysis

Sensitivity reveals which variables are essential for the circuit's proper operation. This analysis establishes the degree to which each component (acting alone or in conjunction with other components) influences the operation of the circuit. Additionally, it modifies every value to simulate the worst-case scenario. The sensitive components are identified using this analysis, and the results are exported to the Optimizer to calibrate the behavior of the circuit. Sensitivity analysis is also used to identify the components that have an impact on production, after which the tolerance of the sensitive components is decreased and that of the less sensitive components is increased.

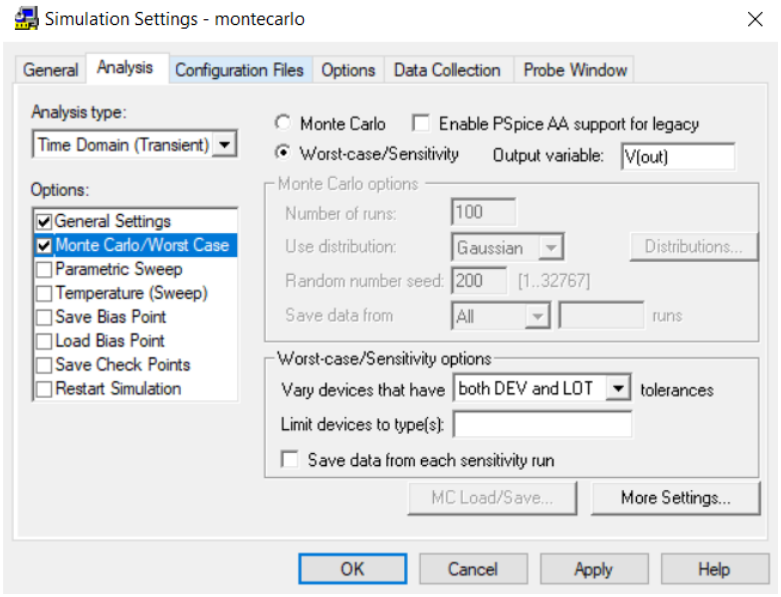


Figure 21 The Worst Case/Sensitivity analysis simulation profile

In the figure below (Figure 21) we can observe the information provided by the worst case analysis about what components are the most sensitive and critical to the correct functioning of the system, and also remark what component has the highest deviation form its nominal value.

```
                    WORST CASE ALL DEVICES

*******************************************************************************


Device         MODEL          PARAMETER     NEW VALUE
C_C            C_C            C                 .8          (Decreased)
R_R1           R_R1           R                1.05         (Increased)
R_R2           R_R2           R                 .95         (Decreased)
R_Rp           R_Rp           R                 .95         (Decreased)

**** 05/20/23 16:52:15 ****** PSpice 17.2.0 (March 2016) ****** ID# 0 ********

 ** Profile: "SCHEMATIC1-montecarlo"  [ d:\utcn\teme\an ii\cad\project\bun-pspicefiles\schematic1\montecarlo.sim ]

 ****     SORTED DEVIATIONS OF V(OUT)      TEMPERATURE =   27.000 DEG C

                    WORST CASE SUMMARY

*******************************************************************************


Mean Deviation =     .6798
Sigma          =    0

 RUN                    MAX DEVIATION FROM NOMINAL

WORST CASE ALL DEVICES
                         .6798  higher  at T =    3.3783E-03
                        ( 107.24% of Nominal)
```

Figure 22 The worst-case analysis results

4. Bill of materials

Quantity	Catalog Number	Description	Price ($)
1	3757-MMSZ5244A_R1_00001CT-ND	SOD-123, ZENER	0,19
4	1N4148FSCT-ND	DIODE GEN PURP 100V 200MA DO35	0,10
1	497-10230-1-ND	IC OPAMP JFET 2 CIRCUIT 8SOIC	0,86
2	2383-2N2222-ND	TRANS NPN 30V 0.8A TO18	0,50
1	2N3055MS-ND	TRANS NPN 70V 15A TO3	1,50
2	CF14JT1K00CT-ND	RES 1K OHM 5% 1/4W AXIAL	0,10
1	749-AC05000005001JAC00CT-ND	RES 5K OHM 5W 5% AXIAL	1,07
1	P160KNP-0EC15B5K-ND	POT 5K OHM 1/5W PLASTIC LINEAR	0,69084
1	PNP1WVFT-52-0R25-ND	RES 0.25 OHM 1% 1W AXIAL	0,24684
1	1189-1125-ND	CAP ALUM 10000UF 20% 16V RADIAL	2,14

5. Conclusion

In conclusion, the voltage regulator project was successfully designed and implemented within the specified voltage range. By carefully selecting and configuring the appropriate components, the circuit was able to maintain a constant output voltage regardless of changes in input voltage or load conditions. Through the utilization of OrCAD, a powerful circuit simulation tool, the design and analysis of the voltage regulator were made efficient and accurate.

With the successful implementation of the voltage regulator and the assistance of OrCAD, we have achieved a reliable and robust solution for regulating voltages within the desired range. This project has not only deepened our understanding of voltage regulation principles but also highlighted the importance of utilizing advanced simulation tools like OrCAD for efficient circuit design and analysis.

6. Bibliography

- https://engineeringtutorial.com/basic-dc-power-supply-circuit/
- https://www.britannica.com/technology/transformer-electronics
- https://www.tutorialspoint.com/electronic_circuits/electronic_circuits_filters.htm
- http://www.bel.utcluj.ro/dce/didactic/fec/16_voltage_regulators.pdf
- https://www.researchgate.net/publication/352778322_Generation_of_Control_Signals_for_an_AC_-DC_Converters_for_a_Hybrid_Charging_System
- https://www.electronics-tutorials.ws/diode/diode_6.html
- CAD Techniques laboratory notes